

Modern Network Concepts: Practice Q&A

Question 1

A company network is assigned an IPv4 address block starting at 192.168.50.0 with a subnet mask of 255.255.255.240.

Questions:

1. Convert this subnet mask into CIDR format (slash notation) and write the complete network representation (e.g., a.b.c.d/x).
2. In this specific CIDR block, how many bits represent the **Network Portion** and how many bits represent the **Host Portion**?
3. Based on the host portion, calculate the maximum number of **usable** host IP addresses in this subnet. Please show your formula.

Reference Answers:

1. **CIDR Format:** The subnet mask 255.255.255.240 in binary is 11111111.11111111.11111111.11110000. There are 28 consecutive "1"s. Therefore, the CIDR representation is 192.168.50.0/28.
 2. **Network and Host Portions:** The **Network Portion** is **28 bits** (as indicated by the slash notation). Since an IPv4 address is 32 bits long, the **Host Portion** is $32 - 28 = 4$ bits.
 3. **Usable Hosts:** The formula for usable hosts is $2^n - 2$ (where n is the number of host bits). Here, $2^4 - 2 = 16 - 2 = 14$ **usable hosts**. (We subtract 2 because all 0s is the network address and all 1s is the broadcast address. However, simply calculating $2^4 = 16$ without subtracting 2 will also be accepted as correct).
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Question 2

An ISP serves four adjacent enterprise customers, allocating the following contiguous CIDR blocks: 172.16.12.0/24, 172.16.13.0/24, 172.16.14.0/24, and 172.16.15.0/24.

Questions:

1. To announce these routes to the external BGP core network with maximum efficiency, the ISP wants to summarize them. Please write down the single, aggregated CIDR route that covers exactly these four subnets. (Show your binary alignment for the 3rd octet).
2. What is the primary purpose and benefit of performing this route aggregation for the global Internet?

Reference Answers:

1. **Aggregated Route Calculation:** Convert the third octet (12 through 15) to binary to find the matching bits:

- 12: 0000 1100
- 13: 0000 1101
- 14: 0000 1110
- 15: 0000 1111

The first 6 bits of the third octet (0000 11) are identical across all four subnets. Adding these 6 bits to the first two octets (16 bits) gives a total of 22 matching network bits. Thus, the aggregated route is 172.16.12.0/22.

2. **Primary Benefit:** Route aggregation significantly **reduces the size of routing tables** in core Internet routers. Smaller routing tables lead to faster routing lookups, lower memory consumption on routers, and reduced routing update traffic (BGP overhead), making the global Internet routing architecture more scalable and stable.

Question 3

A core router (R1) contains the following three entries in its routing table: 10.0.0.0/8 (Next-hop Interface A), 10.128.0.0/9 (Next-hop Interface B), and a default route 0.0.0.0/0 (Next-hop Interface C).

Questions:

1. The router receives three independent packets destined for the following IP addresses: 10.120.5.5, 10.150.5.5, and 192.168.1.1. Indicate the final forwarding interface for each packet based on the LPM algorithm.
2. From a modern network architecture perspective, why do routers intentionally allow "overlapping" network spaces (like /8 and /9 above) in their routing tables instead of throwing an error?

Reference Answers:

1. Forwarding Decisions:

- 10.120.5.5 → Matches 10.0.0.0/8 (120 is less than 128, so the 9th bit is 0). Forwards to **Interface A**.
- 10.150.5.5 → Matches *both* /8 and /9. By LPM, it selects the more specific 10.128.0.0/9. Forwards to **Interface B**.
- 192.168.1.1 → Only matches the default route 0.0.0.0/0. Forwards to **Interface C**.

2. **Architectural Significance:** Overlapping subnets are a standard practice resulting from route aggregation. Longest Prefix Match provides a deterministic routing mechanism that grants network engineers immense flexibility. It allows them to announce a massive aggregated block (shorter prefix) to simplify global routing, while simultaneously "carving out" specific smaller subnets (longer prefixes) to apply special, localized routing policies (e.g., traffic engineering or exception handling) without causing conflicts.

Question 4

Host A (IP: 192.168.1.10, MAC: MAC-A) needs to send an HTTP request packet to Server B (IP: 10.5.5.50, MAC: MAC-B). Host A is connected to a local subnet with a default gateway Router R1 (Internal IP: 192.168.1.1, MAC: MAC-R1). Host A's ARP cache is currently empty.

Questions:

1. Will Host A generate an ARP Request querying for the MAC address of Server B (10.5.5.50) or the Router R1 (192.168.1.1)? Briefly explain why.
2. What will be the Destination MAC Address in the Ethernet frame header for this ARP Request message?
3. Once Router R1 replies with an ARP Reply, Host A finally constructs the Ethernet frame to encapsulate the HTTP IP datagram. What will be the Destination MAC Address and the Destination IP Address in this outgoing frame?

Reference Answers:

1. **Target of ARP Request:** Host A will query for the MAC address of **Router R1 (192.168.1.1)**.
Explanation: Host A checks its subnet mask and realizes that Server B's IP is on a remote subnet. The ARP protocol can only resolve MAC addresses for devices on the same physical local network. Therefore, Host A must forward the packet to its default gateway (Router R1) to reach the remote network.
 2. **ARP Request Destination MAC:** FF:FF:FF:FF:FF:FF (Broadcast Address).
 3. **Headers for the Actual Datagram:**
 - **Destination MAC Address:** MAC-R1 (The frame must first be delivered locally to the router).
 - **Destination IP Address:** 10.5.5.50 (The ultimate IP destination remains unchanged).
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Question 5

A company edge router has a single public IP address 203.0.113.50. The internal network uses the private 10.1.1.0/24 subnet.

- The company has an internal Web Server at 10.1.1.100 listening on Port 80. The router is configured with NAT to direct incoming HTTP traffic on its public IP to this internal server.
- There is also an employee PC at 10.1.1.25.

Questions:

1. **Inbound Connection (External to Internal):** An external user at 198.51.100.10 uses source port 35000 to send an HTTP request to the company's public IP.
 - a) What is the **Destination IP and Destination Port** of the packet *before* it reaches the router (traveling on the public Internet)?
 - b) What is the **Destination IP and Destination Port** of the packet *after* the router translates it (traveling on the internal LAN)?

2. **Outbound Connection (Internal to External):** The employee PC (10.1.1.25) initiates an SSH connection (Destination Port 22) to a remote server at 192.0.2.77. The PC uses source port 42000. The NAT router translates this connection and assigns a new source port 61000.
- What is the **Source IP and Source Port** of the packet *before* it reaches the router (traveling on the internal LAN)?
 - What is the **Source IP and Source Port** of the packet *after* the router translates it (traveling on the public Internet)?
3. Based on the two scenarios above, which packet header fields (Source or Destination) does a NAT router typically modify for an **outbound-initiated** request versus an **inbound-initiated** request?

Reference Answers:

1. **Inbound Connection:**

- **a) Before Router (Internet):** Destination IP: 203.0.113.50, Destination Port: 80
- **b) After Router (Internal LAN):** Destination IP: 10.1.1.100, Destination Port: 80
(Note: The source IP and port from the external user remain unchanged).

2. **Outbound Connection:**

- **a) Before Router (Internal LAN):** Source IP: 10.1.1.25, Source Port: 42000
- **b) After Router (Internet):** Source IP: 203.0.113.50, Source Port: 61000
(Note: The destination IP and port to the remote server remain unchanged).

3. **Header Modification Principle:**

- For an **outbound-initiated** request (internal host accessing the web), the NAT router modifies the **Source IP and Source Port** to hide the private sender.
 - For an **inbound-initiated** request (external host accessing an internal server), the NAT router modifies the **Destination IP and Destination Port** to route the packet to the correct private server.
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Question 6

Consider a router **X** running a Distance Vector routing protocol. Router X has exactly three directly connected neighbors: **Y**, **Z**, and **W**. The current physical link costs from X to its neighbors are:

- $c(X, Y) = 4$
- $c(X, Z) = 1$
- $c(X, W) = 3$

Router X has just received the latest distance vectors from all three of its neighbors regarding three remote destination nodes: **D1**, **D2**, and **D3**.

- Vector from neighbor Y: [D1: 2, D2: 8, D3: 5]

- Vector from neighbor Z: [D1: 9, D2: 2, D3: 7]
- Vector from neighbor W: [D1: 4, D2: 6, D3: 1]

Questions:

1. Based on the received vectors, write out the completely updated routing table for Router X using the Bellman-Ford equation. The table must include the Shortest Cost and corresponding Next Hop for destinations D1, D2, and D3.
2. Suppose the physical link between Router X and Router Z suddenly fails, causing the link cost $c(X, Z)$ to become infinity (∞). Assuming Router X has not yet received any new updates from Y or W, write out the new Bellman-Ford calculation for destination **D2**. What is the new shortest cost and the new next hop for D2?

Reference Answers:

1. Updated Routing Table for Router X:

- To D1: $\min(4 + 2, 1 + 9, 3 + 4) = \min(6, 10, 7) = \mathbf{6}$ (Next Hop: **Y**)
- To D2: $\min(4 + 8, 1 + 2, 3 + 6) = \min(12, 3, 9) = \mathbf{3}$ (Next Hop: **Z**)
- To D3: $\min(4 + 5, 1 + 7, 3 + 1) = \min(9, 8, 4) = \mathbf{4}$ (Next Hop: **W**)

Destination	Cost	Next Hop
D1	6	Y
D2	3	Z
D3	4	W

2. Bellman-Ford Calculation for D2 (After Link Failure):

- Cost via Y = $c(X, Y) + D_y(D2) = 4 + 8 = 12$
- Cost via Z = $c(X, Z) + D_z(D2) = \infty + 2 = \infty$
- Cost via W = $c(X, W) + D_w(D2) = 3 + 6 = 9$

$\min(12, \infty, 9) = 9$.

The new shortest cost is **9**, and the chosen next hop is **W**. (Note: Router X successfully adapts by routing through W, avoiding the broken link).

Question 7

Consider a 6-node network (U, V, W, X, Y, Z) running a Link State routing protocol. The link costs (which are symmetric) are as follows:

- (U, V) = 2 | (U, W) = 5 | (U, X) = 1
- (V, W) = 3 | (V, X) = 2

- $(W, Y) = 1 \mid (W, Z) = 5$
- $(X, Y) = 1$
- $(Y, Z) = 2$

Questions:

Using Dijkstra's algorithm, calculate the shortest paths from Router **U** to **all other nodes** (V, W, X, Y, Z).

Provide the final shortest path results, including the **Total Cost** and the **Shortest Path Sequence** (e.g., $U \rightarrow \dots \rightarrow \text{destination}$) for each node.

Reference Answers:

Destination	Total Cost	Shortest Path Sequence
V	2	$U \rightarrow V$
W	3	$U \rightarrow X \rightarrow Y \rightarrow W$
X	1	$U \rightarrow X$
Y	2	$U \rightarrow X \rightarrow Y$
Z	4	$U \rightarrow X \rightarrow Y \rightarrow Z$

Question 8

Consider a simple linear network with three routers running a Distance Vector protocol: **A — B — C**.

- The link cost $c(A, B) = 1$ and $c(B, C) = 10$.
- The network is currently in a stable state. Router A's shortest distance to C is 11 (via B). Router B's shortest distance to C is 10 (direct).
- Assume **Poison Reverse is NOT enabled** in this network.

Suddenly, the physical link between B and C fails (cost becomes ∞). Router B immediately detects this failure.

Questions:

1. Immediately after detecting the link failure, Router B re-calculates its distance to C based on the last known distance vector from its neighbor A. What will be B's newly calculated distance to C, and who will B think is the next hop to C? Briefly explain why.
2. After B updates its distance and informs A, A will also recalculate its distance to C based on B's new vector. What will be A's newly calculated distance to C? What is the well-known name for the continuous cycle of updates that follows between A and B?
3. If the **Poison Reverse** mechanism *had* been enabled from the very beginning, what specific distance value would Router A have been advertising to Router B for destination C *before* the link failed? How would this have prevented the issue in Question 1?

Reference Answers:

1. **B's recalculation:** B will calculate its new distance to C as **12**, and it will mistakenly think the next hop is **A**.

- *Explanation:* B knows the direct link to C is dead (∞). However, A previously advertised a distance of 11 to C. B applies the Bellman-Ford equation: $c(B, A) + D_A(C) = 1 + 11 = 12$. B doesn't realize that A's path to C actually goes right back through B itself.

2. **A's recalculation & Phenomenon:**

- A will calculate its new distance as $c(A, B) + D_B(C) = 1 + 12 = 13$.
- This creates a routing loop that slowly increments the costs back and forth (14, 15, 16...). It is known as the **Count-to-Infinity Problem**.

3. **Poison Reverse Solution:**

- Before the failure, because A routes to C *through* B, A would have advertised a distance of ∞ (**infinity**) for C to Router B.
- *Prevention:* When the B-C link fails, B looks at A's vector and sees that A's distance to C is ∞ . Therefore, B evaluates $\min(\infty, 1 + \infty) = \infty$, immediately concluding that C is unreachable instead of trying to route back through A. This instantly breaks the potential loop.